

Cross-Institution Generalization of Deep Learning Skin Cancer Classifiers

(2022)

Problem

Understanding why skin cancer classifiers trained at one institution fail to generalize elsewhere.

Method

A ResNet-50 baseline benchmarked across institutions, plus domain adaptation experiments.

Dataset

Multi-institution dermoscopic image collections.

Evaluation

AUC drop when testing across different collection sites.

Results

AUC dropped from 0.93 in-domain to 0.79 on an external site, even after standard domain adaptation.

Limitations

The generalization gap remains largely unresolved; explicitly called out as an open problem in dermatology AI.